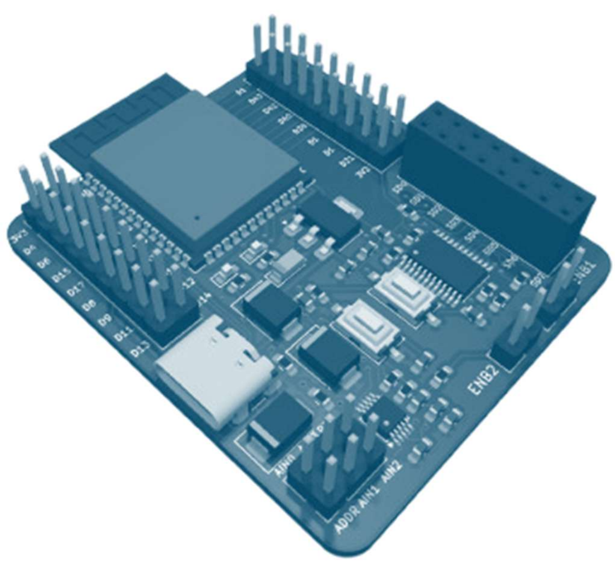
BIOMED-S3

Custom ESP32-S3 Biomedical Microcontroller Board

Datasheet | Revision 1.0

April 2026

Core	Connectivity	Sensors	Interface
ESP32-S3-WROOM-1 16MB Flash / 8MB PSRAM	Wi-Fi 802.11 b/g/n Bluetooth 5 / BLE	TCA9548A I2C Mux ADS1115 16-bit ADC	USB-C OTG GPIO Headers



1. Overview

The BIOMED-S3 is a compact, high-performance biomedical microcontroller board built around the Espressif ESP32-S3-WROOM-1 module. It is designed for medical-grade sensor interfacing, data acquisition, and wireless transmission in portable and bench-top biomedical applications, which is created by Jewzon Ravindra.

The board integrates the TCA9548APWR 8-channel I2C multiplexer and ADS1115IDGS 16-bit ADC, allowing connection of multiple I2C sensors simultaneously with precision analog front-end capability. Address configuration is user-selectable via onboard jumper pads.

Key Features

- ESP32-S3-WROOM-1 — Dual-core Xtensa LX7 @ 240 MHz, 16 MB Flash, 8 MB PSRAM
- Wi-Fi 802.11 b/g/n (2.4 GHz) + Bluetooth 5.0 / BLE
- USB-C OTG port (GPIO19/GPIO20) — firmware flashing, CDC serial, HID
- TCA9548APWR — 8-channel I2C switch, address configurable via jumper pads
- ADS1115IDGS — 16-bit, 4-channel ADC with PGA, address configurable via jumper pads
- AMS1117-3.3 LDO voltage regulator
- BOOT and RESET buttons
- Power and user LEDs
- Exposed GPIO headers — all available GPIOs broken out

2. Electrical Specifications

2.1 Absolute Maximum Ratings

Exceeding these values may permanently damage the device.

Parameter	Symbol	Min	Max	Unit
Supply voltage (VIN)	VIN	-0.3	6.0	V
3.3V rail voltage	VDD	-0.3	3.6	V
GPIO pin voltage	VGPIO	-0.3	3.6	V
Storage temperature	TSTG	-40	+125	°C

2.2 Recommended Operating Conditions

Parameter	Min	Typ	Max	Unit
Input supply voltage (USB-C)	4.5	5.0	5.5	V
3.3V output (AMS1117)	3.235	3.3	3.365	V
Operating temperature	-20	25	+85	°C
Operating humidity (non-condensing)	10	—	90	% RH
Maximum current (3.3V rail)	—	—	800	mA

2.3 Power Consumption

Mode	Typical Current	Notes
Active (Wi-Fi TX)	~240 mA	Peak during transmission
Active (CPU only)	~80 mA	No RF, full clock
Modem Sleep	~20 mA	CPU active, Wi-Fi off
Deep Sleep	~10 µA	RTC only

3. Pin Description

3.1 USB-C Connector (J1)

Pin	Name	Type	Description
A4/B4	VBUS	Power	5V USB supply input
A5/B5	CC1/CC2	Config	5.1kΩ pull-down for USB-C device mode
A6	D+	USB Data	Connected to GPIO20 (USB_D+)
A7	D-	USB Data	Connected to GPIO19 (USB_D-)
A1/B1	GND	Ground	Shield and signal ground

3.2 GPIO Header Pinout (J2 & J5)

All available GPIOs from the ESP32-S3-WROOM-1 module are exposed on dual-row headers. Pins reserved for SPI Flash/PSRAM (GPIO35–37) are not available for user I/O.

GPIO	Alt Function	Direction	Notes
GPIO0	BOOT	Input	Strapping pin — hold LOW for download mode
GPIO1–7	ADC / Touch	I/O	General purpose, Analog capable
GPIO8	I2C SDA	I/O	Default I2C data — TCA9548A & ADS1115
GPIO9	I2C SCL	I/O	Default I2C clock — TCA9548A & ADS1115
GPIO19	USB_D-	USB	USB OTG D- — do not use for GPIO when USB active
GPIO20	USB_D+	USB	USB OTG D+ — do not use for GPIO when USB active
GPIO43	U0TXD	Output	UART0 TX — debug serial
GPIO44	U0RXD	Input	UART0 RX — debug serial
GPIO45	VSPI	Strapping	Strapping pin — avoid external pull on boot
GPIO46	—	Strapping	Strapping pin — avoid external pull on boot
ENB1	Enable pin	Selector	To enable TCA9548A
ENB2	Enable pin	Selector	To enable ADS1115

4. Onboard Peripherals

4.1 TCA9548APWR — 8-Channel I2C Multiplexer

The TCA9548APWR provides 8 independent I2C channels, enabling connection of multiple sensors sharing the same I2C address without conflicts. It is connected to the ESP32-S3 via the default I2C bus (GPIO8/GPIO9).

Parameter	Value	Notes
Default I2C Address	0x70	A2=A1=A0=GND
Address Range	0x70 – 0x77	Set via jumper pads A0, A1, A2
Number of Channels	8 (SC0–SC7 / SD0–SD7)	Software selectable
Supply Voltage	3.3V	Powered from onboard 3.3V rail
I2C Speed	Up to 400 kHz	Fast mode supported
Enable Pin	ENB2	By Default it's Enabled

4.1.1 TCA9548A Address Jumper Pads

Three solder jumper pads (A0, A1, A2) on the bottom of the PCB control the I2C address. Bridge a pad to connect that address pin to VCC (logic HIGH). Leave open for GND (logic LOW).

A2	A1	A0	I2C Address
Open (0)	Open (0)	Open (0)	0x70 (Default)
Open (0)	Open (0)	Bridged (1)	0x71
Open (0)	Bridged (1)	Open (0)	0x72
Bridged (1)	Open (0)	Open (0)	0x74
Bridged (1)	Bridged (1)	Bridged (1)	0x77

4.2 ADS1115IDGS — 16-Bit ADC

The ADS1115IDGS is a precision, low-power, 16-bit ADC with a programmable gain amplifier (PGA) and 4 single-ended or 2 differential input channels. It communicates via I2C and is suitable for ECG, EMG, temperature, and pressure sensor acquisition.

Parameter	Value	Notes
Resolution	16-bit	±32768 counts
Input Channels	4 single-ended / 2 differential	AIN0–AIN3
PGA Range	±0.256V to ±6.144V	6 gain settings
Data Rate	8 – 860 SPS	Software configurable
Default I2C Address	0x48	ADDR pin = GND
Address Range	0x48 – 0x4B	Set via ADDR jumper pad
Supply Voltage	3.3V	Powered from onboard 3.3V rail
Enable Pin	ENB1	By Default it's Enabled

4.2.1 ADS1115 Address Jumper Pad

A single ADDR solder jumper pad selects the I2C address of the ADS1115 by connecting the ADDR pin to GND, VDD, SDA, or SCL.

ADDR Jumper Connection	I2C Address
GND (Default — open pad)	0x48
VDD (Bridge to 3.3V)	0x49
SDA (Bridge to SDA line)	0x4A
SCL (Bridge to SCL line)	0x4B

4.3 Power Regulation — AMS1117-3.3

The AMS1117-3.3 LDO linear regulator converts the 5V USB-C input to a stable 3.3V supply for the ESP32-S3 module and all onboard peripherals.

Parameter	Value	Notes
Input Voltage	4.75V – 5.5V	From USB VBUS
Output Voltage	3.3V ± 1%	Fixed
Max Output Current	800 mA	With adequate heatsinking
Dropout Voltage	~1.1V @ 800mA	

5. USB-C OTG Interface

The board exposes the ESP32-S3 native USB OTG interface directly via the USB-C connector. No USB-to-UART bridge chip is used on this port — the ESP32-S3 USB PHY is connected directly.

Feature	Detail
USB Standard	USB 2.0 Full Speed (12 Mbps)
D+ Pin	GPIO20
D- Pin	GPIO19
Supported Classes	CDC-ACM (Virtual COM), HID, DFU, MSC
CC Resistors	5.1kΩ pull-down on CC1 and CC2 (device mode)
ESD Protection	None onboard — add USBLC6-2SC6 for production
Firmware Requirement	USB Mode must be set to USB-OTG (TinyUSB) in Arduino IDE

5.1 Arduino IDE Configuration for USB OTG

- Board: ESP32S3 Dev Module
- USB Mode: USB-OTG (TinyUSB)
- USB CDC On Boot: Enabled
- Flash Size: 16MB
- Partition Scheme: 16M Flash (3MB APP / 9.9MB FATFS)

5.2 Entering Download Mode

To flash firmware via USB or UART:

- Hold the BOOT button (GPIO0 LOW)
- Press and release the RESET button
- Release the BOOT button
- The board is now in download mode — proceed with flashing

6. I2C Bus Architecture

The board uses a single I2C master bus from the ESP32-S3 connected to both the TCA9548A multiplexer and the ADS1115 ADC. The TCA9548A enables expansion to 8 downstream I2C channels for additional sensor integration.

Signal	ESP32-S3 Pin	Pull-up	Notes
SDA	GPIO8	4.7kΩ to 3.3V	Shared bus
SCL	GPIO9	4.7kΩ to 3.3V	Shared bus

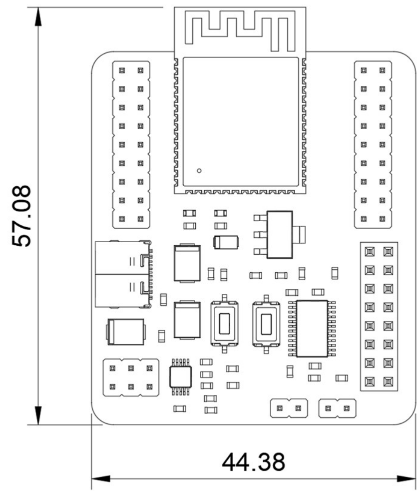
Default I2C Device Address Map

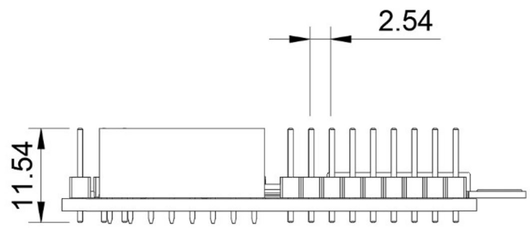
Device	Default Address	Configurable?	Via
TCA9548APWR	0x70	Yes (0x70–0x77)	Jumper pads A0–A2
ADS1115IDGS	0x48	Yes (0x48–0x4B)	Jumper pad ADDR

7. Mechanical Specifications

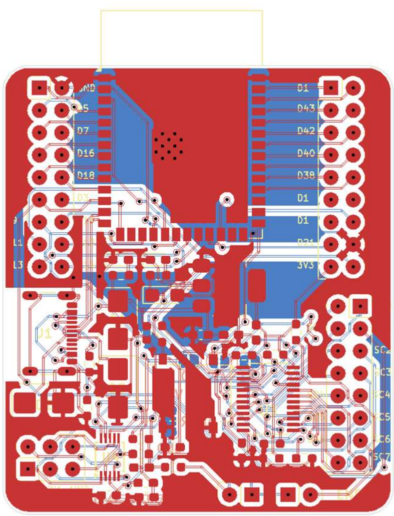
Parameter	Value
Board Form Factor	Custom PCB
PCB Layers	2-layer (recommended 4-layer for production)
PCB Thickness	1.6 mm
Connector	USB Type-C Receptacle (16P)
GPIO Header Pitch	2.54 mm (0.1")
Operating Temperature	-20°C to +85°C

7.1 Dimensions

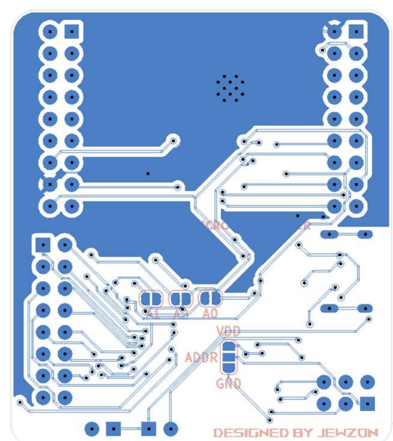




7.2 PCB Design



Front & Back



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7.3 Actual Picture

8. Quick Start Guide

8.1 Hardware Setup

- Connect the board to your PC using a USB-C data cable
- Ensure the cable supports data (not charge-only)
- The PWR LED should illuminate when powered
- For UART flashing, connect a USB-UART adapter to GPIO43 (TX) and GPIO44 (RX)

8.2 Arduino IDE Setup

- Install ESP32 board package by Espressif (v2.x or later) via Boards Manager
- Select: Tools → Board → ESP32S3 Dev Module
- Set USB Mode → USB-OTG (TinyUSB)
- Set Flash Size → 16MB
- Select the correct COM port

8.3 Reading ADS1115 (Example)

Install the Adafruit ADS1X15 library from Library Manager, then:

```
#include <Adafruit_ADS1X15.h>
Adafruit_ADS1115 ads;
void setup() { Wire.begin(8,9); ads.begin(0x48); }
void loop() { int16_t v = ads.readADC_SingleEnded(0); }
```

9. Safety & Regulatory Notice

This board is a development and evaluation module. It has not been certified under any medical device regulatory framework (FDA, CE, MDR, etc.). It is NOT approved for direct clinical use or patient-connected applications without additional safety evaluation, isolation circuitry, and regulatory approval.

- For patient-connected biomedical applications, ensure adequate galvanic isolation
- USB-C port shares ground with all GPIO — use optical or transformer isolation for ECG/EEG front-ends
- The ADS1115 input range must not exceed ±VDD when using external sensors
- ESD protection on analog input lines is recommended for production designs

10. Revision History

Rev	Date	Author	Changes
1.0	April 2026	—	Initial release

11. Reference

For more information kindly visit: Espressif Centralized Documentation Platform (CDP) | Espressif Documentation